End-sem: worked solutions

25 Nov 2024 · 50 marks · 3 hours · CEPC12

The paper does not print a section. The Section B (Batch B) paper of the same date is separate, so this one is filed as Section A.

Question 1

Q1 (a) — 2 marks

> Explain surface tension and capillarity.

- **Surface tension (σ)**: the tensile force per unit length acting along the free surface of a liquid, caused by **cohesion**: a molecule at the surface has no liquid above it, so it is pulled inward, and the surface behaves like a stretched membrane (N/m; water ≈ 0.073 N/m). It explains spherical droplets, the excess pressure inside drops ($p = 4\sigma/d$) and bubbles ($p = 8\sigma/d$).
- **Capillarity**: the **rise or fall of a liquid in a narrow tube**, caused by surface tension together with adhesion and cohesion. If adhesion > cohesion (water–glass) the liquid **rises**; if cohesion > adhesion (mercury) it is **depressed**. $h = 4\sigma\cos\theta/(\rho g d)$.

Q1 (b) — 4 marks

> A caisson closing the entrance to a dry dock is trapezoidal, 16 m wide at the top, 10 m wide at the bottom and 6 m deep. Find the total pressure and centre of pressure if the water outside is level with the top and the dock is empty.

At depth y below the top, the width is $b(y) = 16 - y$.

$A = (16 + 10)/2 \times 6 = 78\ m^2$

Centroid depth $\bar{h} = (h/3)(a + 2b)/(a + b) = (6/3)(16 + 20)/26 = 2.769$ m

$F = \rho g A \bar{h} = 9810 \times 78 \times 2.769 =$ **2119 kN**

Centre of pressure by integration (second moment about the free surface):

$I_O = \int_0^6 y^2 (16 - y)\, dy = 16 \times 72 - 6^4/4 = 1152 - 324 = 828\ m^4$

$h^* = I_O / (A \bar{h}) = 828 / (78 \times 2.769) =$ **3.833 m below the water surface**

Q1 (c)

4 marks

> An inverted differential manometer is connected to pipes A and B conveying water; the manometer fluid is oil of S = 0.8. Find the pressure difference between A and B for the readings shown.

From the figure: X–X is the oil–water interface in the left limb; A is 30 cm below X–X. In the right limb the oil extends 30 cm below X–X, and B is a further 20 cm below that interface.

Left limb at X–X: $p_A - 1000 \times 9.81 \times 0.3 = p_A - 2943$

Right limb at X–X: $p_B - 1000 \times 9.81 \times 0.2 - 800 \times 9.81 \times 0.3 = p_B - 1962 - 2354.4 = p_B - 4316.4$

Equating: **$p_B - p_A = 1373.4\ N/m^2$** (B is at the higher pressure)

Question 2

Q2 (a)

2 marks

> Write the moment of inertia about the centroid for: (i) rectangle (ii) triangle (iii) circle (iv) trapezium.

Shape	I_G (about centroidal axis parallel to the base)	Centroid from base
Rectangle (b × d)	$bd^3/12$	d/2
Triangle (base b, height h)	$bh^3/36$	h/3
Circle (diameter d)	$\pi d^4/64$	d/2

Shape	I_G (about centroidal axis parallel to the base)	Centroid from base
Trapezium (parallel sides a, b; height h)	$h^3(a^2 + 4ab + b^2) / [36(a + b)]$	$h(2a + b)/[3(a + b)]$ from side b

Q2 (b)

4 marks

> A solid cylinder 15 cm in diameter and 60 cm long consists of two parts: the base part is 1.20 cm long with S = 5.0; the rest has S = 0.6. Can it float vertically in water?

Work per unit cross-sectional area (lengths in cm, $\rho_w = 1$):

Weight $\propto 5.0 \times 1.2 + 0.6 \times 58.8 = 6.0 + 35.28 = 41.28 \rightarrow$ depth of immersion h = **41.28 cm**

KB = 41.28/2 = 20.64 cm

KG = $(6.0 \times 0.6 + 35.28 \times (1.2 + 29.4)) / 41.28 = (3.6 + 1079.6)/41.28 = 26.24$ cm → BG = 5.60 cm

BM = $I/V = (\pi/64\, d^4)/(\pi/4\, d^2\, h) = d^2/(16h) = 15^2/(16 \times 41.28) = 0.34$ cm

GM = BM − BG = 0.34 − 5.60 = **−5.26 cm**

GM is **negative**, so the cylinder **cannot float vertically**; it is unstable in that position.

Q2 (c)

4 marks

> Determine the third velocity component so that continuity is satisfied: (i) $u = x^2 + y^2 + z^2$, $v = xy^2 - yz^2 + xy$; (ii) $v = 2y^2$, $w = 2xyz$.

Continuity (incompressible): $\partial u/\partial x + \partial v/\partial y + \partial w/\partial z = 0$.

(i) $\partial u/\partial x = 2x$; $\partial v/\partial y = 2xy - z^2 + x$

$\partial w/\partial z = -(2x + 2xy - z^2 + x) = -3x - 2xy + z^2$

$w = -3xz - 2xyz + z^3/3 + f(x, y)$

(ii) $\partial v/\partial y = 4y$; $\partial w/\partial z = 2xy$

$\partial u/\partial x = -(4y + 2xy)$

$$\boxed{u = -4xy - x^2y + f(y, z)}$$

(The arbitrary functions can be taken as zero for the simplest solution.)

Question 3

Q3 (a)

2 marks

> Define flow net and equipotential lines.

- **Equipotential line**: a line along which the **velocity potential ϕ is constant**. There is no flow along it ($d\phi = 0$), and it is everywhere **perpendicular to the streamlines**.
- **Flow net**: a grid formed by a family of **streamlines and equipotential lines** intersecting at right angles, forming approximately square elements. It is used for 2-D irrotational flow (e.g. seepage under dams) to find velocities, discharge and pressures. Only valid for steady, irrotational flow.

Q3 (b)

4 marks

> State and derive Bernoulli's equation with assumptions.

Statement: in steady, ideal, incompressible flow along a streamline, the sum of pressure head, velocity head and datum head is constant: $\mathbf{p/\rho g + V^2/2g + z = constant}$.

Derivation (Euler's equation): for an element of area dA and length ds along a streamline at angle θ, net pressure force $-(\partial p/\partial s)\, ds\, dA$, weight component $-\rho g\, dA\, ds \cos\theta$ ($\cos\theta = dz/ds$), and mass × acceleration $\rho\, dA\, ds\, V\, dV/ds$:

$$-(\partial p/\partial s)\, ds\, dA - \rho g\, dA\, dz = \rho\, dA\, ds\, V\, dV/ds \implies dp/\rho + g\, dz + V\, dV = 0$$

Integrating (ρ constant): $p/\rho + gz + V^2/2 = \text{constant} \implies$

$$\boxed{p/\rho g + V^2/2g + z = \text{constant}}$$

Assumptions: ideal (inviscid) fluid; steady; incompressible; irrotational; along a streamline; only gravity and pressure forces.

Q3 (c)

4 marks

> An orifice meter with a 15 cm orifice is inserted in a 30 cm pipe. A mercury–oil differential manometer reads 50 cm of mercury. Find the flow of oil (S = 0.9) if C_d = 0.64.

$h = x(S_{Hg}/S_{oil} - 1) = 0.5 \times (13.6/0.9 - 1) = 7.056$ m of oil

$a_0 = \pi/4 \times 0.15^2 = 0.01767$ m²; $a_1 = \pi/4 \times 0.30^2 = 0.07069$ m²

$Q = C_d\, a_0 a_1 \sqrt{(2gh)}/\sqrt{(a_1^2 - a_0^2)} = 0.64 \times 0.01767 \times 0.07069 \times \sqrt{(2 \times 9.81 \times 7.056)} / \sqrt{(0.07069^2 - 0.01767^2)}$

$Q =$ **$0.1374\ m^3/s \approx 137.4$ l/s**

Question 4

Q4 (a)

2 marks

> What are the hydraulic coefficients of an orifice? Define each.

- **Coefficient of contraction C_c** = area of jet at the vena contracta ÷ area of orifice (0.61–0.69).
- **Coefficient of velocity C_v** = actual velocity at the vena contracta ÷ theoretical velocity $\sqrt{(2gH)}$ (0.95–0.99).
- **Coefficient of discharge C_d** = actual discharge ÷ theoretical discharge = $C_c \times C_v$ (0.61–0.65).
- **Coefficient of resistance C_r** = loss of head in the orifice ÷ head of water.

Q4 (b)

4 marks

> Oil of viscosity 0.1 N·s/m² and RD 0.9 flows through a 50 mm pipe 300 m long at 3.5 l/s. Find the pressure drop over 300 m and the shear stress at the wall.

$V = 0.0035/(\pi/4 \times 0.05^2) = 1.783$ m/s; $Re = 900 \times 1.783 \times 0.05/0.1 = 802 \rightarrow$ **laminar** (< 2000)

Hagen–Poiseuille: $\Delta p = 32\mu VL/D^2 = 32 \times 0.1 \times 1.783 \times 300/0.05^2 =$ **$684{,}500$ N/m² = 68.45 N/cm²**

Wall shear $\tau_0 = (\Delta p/L)(R/2) = (684{,}500/300) \times 0.025/2 =$ **28.5 N/m²**

Q4 (c)

4 marks

> A 300 mm pipe 1000 m long connects two reservoirs with a 15 m level difference. Find the discharge. If an additional 300 mm pipe 600 m long is attached alongside the last 600 m of the existing pipe, find the increase in discharge. f = 0.02; neglect minor losses.

Taking f = 0.02 as the Darcy friction factor ($h_f = fLV^2/2gd$); $A = 0.0707$ m².

Single pipe: $15 = 0.02 \times 1000\, V_1^2/(19.62 \times 0.3) \rightarrow V_1 = 2.101$ m/s $\rightarrow Q_1 =$ **0.1485 m³/s**

With the parallel branch (first 400 m carries Q; last 600 m carries Q/2 in each pipe):

$15 = (0.02/(19.62 \times 0.3))\,[400\, V^2 + 600\, (V/2)^2] = 1.869\, V^2 \rightarrow V = 2.833$ m/s $\rightarrow Q_2 = 0.2003$ m³/s

Increase = **0.0518 m³/s**

(If f is read as the coefficient in $h_f = 4fLV^2/2gd$: $Q_1 = 0.0743$, $Q_2 = 0.1001$, increase 0.0259 m³/s.)

Question 5

Q5 (a)

2 marks

> What is a compound pipe? What is the loss of head when pipes are (i) in series, (ii) in parallel?

A **compound pipe** is a system of two or more pipes of **different diameters or lengths** connected together, in series or in parallel.

- **(i) Series**: the **same discharge** flows through each pipe; the total head loss is the **sum** of the friction losses in each pipe (plus minor losses): $H = \Sigma\, 4f\, L_i V_i^2/(2g\, d_i)$ + minor losses.
- **(ii) Parallel**: the **head loss is the same** in every branch, $h_{f1} = h_{f2} = \ldots$, and the total discharge is the **sum** of the branch discharges, $Q = Q_1 + Q_2 + \ldots$.

Q5 (b)

4 marks

> Define specific speed of a turbine and derive the expression. Explain hydraulic, mechanical and overall efficiency of a turbine.

Specific speed (N_s): the speed of a **geometrically similar turbine** that would develop **1 kW under a head of 1 m**.

Derivation: for similar turbines, $u \propto \sqrt{H}$ and $u = \pi DN/60$, so $\mathbf{N \propto \sqrt{H}/D}$. Discharge $Q \propto (\text{area})(\text{velocity}) \propto D^2\sqrt{H}$. Power $P \propto QH \propto D^2 H^{3/2}$. From $N \propto \sqrt{H}/D$, $D \propto \sqrt{H}/N$, so $P \propto (H/N^2)H^{3/2} = H^{5/2}/N^2 \Longrightarrow N^2 \propto H^{5/2}/P$, i.e. $N = K\, H^{5/4}/\sqrt{P}$. For the specific turbine ($P = 1$, $H = 1$), $N = N_s = K$:

$$\boxed{N_s = N\sqrt{P} \,/\, H^{5/4}} \quad (P \text{ in kW})$$

Pelton: 8.5–30 (one jet); Francis: 50–340; Kaplan: 255–860.

Efficiencies:

- **Hydraulic** η_h = power delivered to the runner ÷ water power at inlet = $(V_{w1}u_1 \pm V_{w2}u_2)/(gH)$.
- **Mechanical** η_m = shaft power ÷ power delivered to the runner.
- **Overall** η_o = shaft power ÷ water power at inlet = $P/(\rho gQH) = \eta_h \times \eta_m$.

Q5 (c)

4 marks

> A Francis turbine with overall efficiency 70% must produce 147.15 kW under 8 m. Peripheral velocity = $0.30\sqrt{(2gH)}$; radial velocity of flow at inlet = $0.96\sqrt{(2gH)}$. Speed 200 rpm; hydraulic losses 20% of available energy. Assuming radial discharge, find (i) guide blade angle, (ii) wheel vane angle at inlet, (iii) wheel diameter at inlet, (iv) width at inlet.

$\sqrt{(2gH)} = \sqrt{(2 \times 9.81 \times 8)} = 12.528$ m/s; $u_1 = 0.30 \times 12.528 = 3.759$ m/s; $V_{f1} = 0.96 \times 12.528 = 12.027$ m/s

$\eta_h = 0.80 = V_{w1}u_1/(gH) \rightarrow V_{w1} = 0.8 \times 9.81 \times 8/3.759 = 16.70$ m/s

(i) $\tan \alpha = V_{f1}/V_{w1} = 12.027/16.70 \rightarrow$ **$\alpha = 35.75°$**

(ii) $\tan \theta = V_{f1}/(V_{w1} - u_1) = 12.027/12.946 \rightarrow$ **$\theta = 42.9°$**

(iii) $D_1 = 60u_1/(\pi N) = 60 \times 3.759/(\pi \times 200) =$ **0.359 m**

(iv) $Q = P/(\eta_o \rho g H) = 147{,}150/(0.7 \times 9810 \times 8) = 2.679$ m^3/s; $B_1 = Q/(\pi D_1 V_{f1}) = 2.679/(\pi \times 0.359 \times 12.027) =$ **0.198 m**